JAMEY MCELVEEN

PRINCIPAL SYSTEMS ARCHITECT | WILEY AUTHOR

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EXECUTIVE SUMMARY

Principal Systems Architect with 30+ years delivering enterprise .NET platforms across SaaS, FinTech, and Healthcare. Designed and sustained multi-tenant systems serving 50,000+ organizations; led teams of 20+ engineers through complete platform generational shifts. Expert in AI-Augmented Engineering: runs LLM-driven tooling (Cursor/Claude) daily to accelerate legacy modernization and reduce friction on complex refactoring work. Published Wiley author. Currently architecting an open-source mission clinic EMR (Christ Medical) with offline-first sync for field clinics in low-connectivity environments.

CORE COMPETENCIES

Languages - C#, .NET 9/Core, T-SQL, TypeScript, C++, Java	**APIs & Protocols** - REST, WCF/SOAP, WS-Security, OpenAPI	**Architecture** - N-Tier, MDD, Strangler, Shard Routing, Offline-First, SOA, Event-Driven
Data - MS SQL, PostgreSQL, Snowflake, Oracle, Redis	**SQL Performance** - Covering Indexes, RowVersion, RLS, Resource Governor	**Cloud & DevOps** - Kubernetes, EKS, Docker, Railway, Vercel, CI/CD
Frontend - Next.js 15, React, Angular 8-15, Storybook, Tailwind, Vue.js	**Observability** - ELK Stack, Structured Logging, OpenTelemetry	**Compliance** - HIPAA, PHI Masking, BAA, TLS 1.2+, FinTech Audit
AI Tooling - Cursor, Claude, LLM Workflows, Clean-Room PII Protocol	**Integration** - Epic EMR, Oracle Cloud, Dotmim Sync, GIS/ESRI	**Leadership** - R&D Mgmt, Mentoring, ADRs, Technical Writing, Scrum Master, Product Owner

PROFESSIONAL EXPERIENCE

Principal Architect, Pro Bono / Kingdom Work

2024 - Present

Christ Medical Mission System | Healthcare / Missions | github.com/christmedical

- Architected offline-first sync layer using Dotmim Sync. Field laptops operate independently and reconcile with the central hub on reconnect. Zero data loss, no paper fallback required.
- Implemented Double Metaphone phonetic patient search via PostgreSQL fuzzystrmatch, solving name romanization inconsistency across multilingual mission contexts.
- Unified spiritual and medical outcome tracking in a single data model, treating the whole person as the patient.
- Enforced CI parity via single gate running ESLint, Vitest, React Testing Library, and production Next.js build locally and in GitHub Actions.

Senior Solutions Architect

2023 - 2025

SecureGive | FinTech / Donation Processing

- Architected the client-facing public API in C# and .NET Core on Kubernetes (EKS), documented with OpenAPI and published at apidocs.securegive.com. Built an Ocelot API gateway for edge protection.
- Designed anti-fraud heuristics to detect and shut down credit-card validation attacks (carding), protecting platform integrity and donor funds.
- Engineered idempotent ETL pipelines for PostgreSQL to Snowflake migrations; pipelines restart mid-stream without duplicating financial records.
- Leveraged Snowflake Time Travel for financial audit trails and year-over-year trend reporting without loading production transactional DB.
- Created, maintained, and documented Angular components using Storybook across the full frontend component library.
- Applied FinTech-grade security including immutable audit trails, TLS 1.2+ enforcement, and least-privilege access across all data surfaces.

Senior Software Developer / Modernization Lead
2021 - 2023

McLeod Health | Enterprise Healthcare / HIPAA

- Migrated entire development organization from file-share-based storage to Git and GitHub, cleaning 100,000+ commits with BFG Repo-Cleaner and reducing repo size by 80%.
- Implemented CI pipelines for HIPAA-compliant clinical systems, replacing manual deployments with automated, auditable release processes.
- Mentored developers and interns on Git workflows, branching strategy, and pull request culture.
- Introduced ELK stack with a Golden Record PHI scrubbing service, replacing real names and SSNs with synthetic identities in non-production environments. Zero BAA exposure risk.
- Enforced TLS 1.2+ across all WCF and REST endpoints.
- Managed data integration bridges between .NET applications and Oracle Cloud / Epic EMR platforms serving thousands of clinical staff.

Programming Manager / R&D Architect
1996 - 2021

ACS Technologies | Enterprise SaaS

Except May 2005 to May 2006 (see Lyle's Data Systems)

- Strategic architectural lead for flagship SaaS suite serving 50,000+ organizations worldwide; directed distributed R&D team of 20+ engineers.
- Scrum Master for Realm for several years; served as Product Owner during management chapter, running backlog grooming, story points, and standups.
- Pioneered Model-Driven Development (MDD) for large-scale C# codebases.
- Used actor pattern for async debit card hardware feedback coordination in payment processing flows.

REALM: MULTI-TENANT SAAS PLATFORM

- Architected multi-tenancy using RLS and SQL Resource Governor to prevent Noisy Neighbor degradation across 50,000+ tenants.
- Built shard-aware routing layer enabling live zero-downtime tenant migration when organizations exceeded resource thresholds.

FACILITY SCHEDULER: N-TIER SCHEDULING ENGINE

- Designed N-tier C# / WCF / SOAP architecture with custom WS-Security header processing in the WCF message pipeline.
- Eliminated high-concurrency booking conflicts using RowVersion optimistic locking; reduced scheduling query overhead 70% via covering index eliminating a bookmark lookup.

HEADMASTER: SCHOOL MANAGEMENT ECOSYSTEM

- Modeled non-linear Student-Guardian-Representative hierarchies using M:N relational tables with bitmask permission flags.

Programming Manager
May 2005 - May 2006

Lyle's Data Systems | Civic Tech / Government SaaS

- Managed three developers delivering Java / Struts MVC government services platforms to municipal clients across the Southeast.
- Architected multi-jurisdiction online property tax payment system enabling citizens and banks to pay home, vehicle, and boat taxes online.
- Designed and built "@ Your Request", a citizen issue-reporting platform with photo uploads and real-time progress tracking.
- Performed GIS / ESRI geospatial data conversion migrating legacy systems into MS SQL Server.

COMMUNITY

FIRST Robotics Technomancers technomancers.github.io

Technical Mentor, International Baccalaureate (IB) STEM Program

- Mentored high school students in Kotlin software development and circuit/sensor integration for competitive robotics.
- Translated complex engineering concepts into accessible instruction for students with no prior programming background.

AI Development, Peer Mentoring

Informal Mentor

- Mentors peers in AI-augmented engineering workflows (Cursor, Claude, prompt engineering, and clean-room PII protocols for regulated environments).
- Extensive self-directed study applied directly to active production projects, not theoretical.

EDUCATION & PUBLICATIONS

B.S. Computer Engineering

Clemson University

C++ / OpenGL background applied to .NET GC optimization and performance-sensitive system design.

Wiley 2009 with Chris Craft

iPhone Game Development

Technical writing clarity applied daily to Swagger/OpenAPI specs, architecture decision records, and deep-dive team documentation.